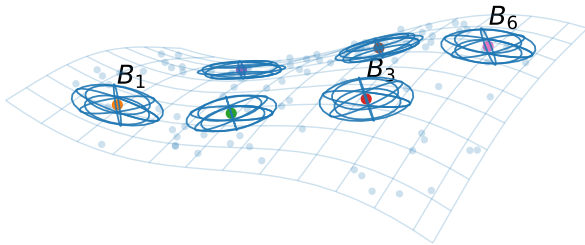


Normality Bubble Diffusion

Source-audited methods and shared MVTec results



Huynh Thanh Dat 14 September 2026

Native methods → Shared MVTec → NBD

One-class detection

Train on normal data

$$\mathcal{D}_N = \{x_i\}_{i=1}^n.$$

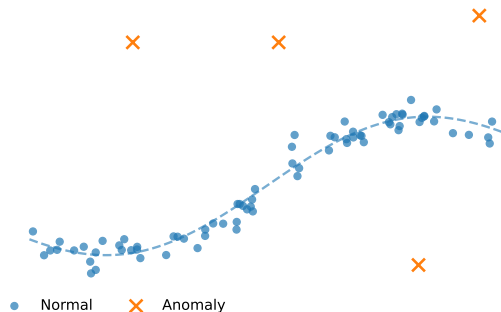
Score new observations

$$s(x) : \mathcal{X} \rightarrow \mathbb{R}.$$

A larger score means more anomalous.

Common MVTec train/calibration:
normal only.

Test: normal + anomaly.



Schematic: normal support can be curved.

Common MVTec uses shared frozen CNN descriptors. Native methods use their own input pipelines.

Two evaluation tracks

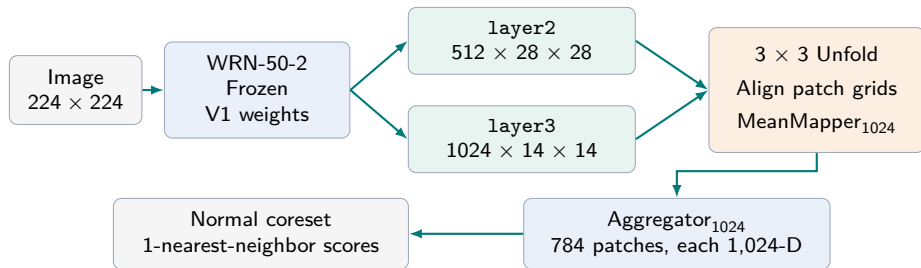
Track	Data	Purpose
Native methods	MNIST / CIFAR / MVTec	Method-specific replication
Common benchmark	MVTec:15 categories	Same features and splits

- User-selected runtime budget: about two hours; training epochs shortened.
- Keep all selected classes, categories and seeds in the coverage manifest.
- Native image CNNs and shared-feature heads are different experiments.

Full data does not mean full paper training schedule.

PatchCore: author feature extractor

Author Quick Guide: ImageNet-pretrained WRN-50 / 224 input / 1,024-D / 10% coreset.



Current rerun: author 1,024-D descriptors. Preserve the startup BN shape probe, then freeze/eval. Native image maximum; common table top-1% mean.

Paper Eq. (7): extra neighbor reweighting. Inspected author code: maximum patch score. Separate targets.

GitHub: [patchcore.py](#) [common.py](#) [Author run command](#)

PatchCore: native MVTec result

Dataset	Variant	Paper (%)	Measured (%)
mvtec	author code	99.00	99.02 $\pm$ 0.14

Pixel AUROC: 98.11%. AUPRO not evaluated.

- Full normal train; 10% coreset; 1NN squared L2; image maximum.
- WR50-2 ImageNet V1; all 15 categories, three seeds.
- Released author code omits paper Eq.(7) reweighting.

Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

Deep SVDD: collapse and safeguards

$$s(x) = \|f_{\theta}(x) - c\|^2, \quad f_{\theta}(x) \equiv c \Rightarrow s(x) = 0.$$

- Train the image CNN end to end; fix a nonzero center c .
- No convolution/linear bias; non-affine BatchNorm; leaky ReLU.
- Reconstruct normal images with an AE, copy its encoder, discard the decoder.
- AE is an initialization; output variance is a collapse diagnostic.

MNIST: conv 8/4 $\rightarrow$ 32. CIFAR: conv 32/64/128 $\rightarrow$ 128. No ImageNet backbone.

Original Theano source | Executed author PyTorch modules.

Deep SVDD: native image CNNs

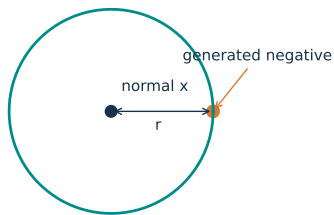
Dataset	Variant	Paper (%)	Measured (%)
cifar10	one-class	64.81	61.09 ± 0.62
cifar10	soft boundary	63.28	61.52 ± 0.63
mnist	one-class	94.80	92.69 ± 0.74
mnist	soft boundary	93.53	92.73 ± 0.78

- Author PyTorch LeNet modules; AE 5 + SVDD 12 epochs.
- Soft-boundary warmup 10; final epoch; 10 classes $\times$ 10 seeds.
- Later author-code replication; original Theano numeric parity unverified.

Paper: Ruff et al., Table 1; class macro of printed means. Protocols differ.

Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

DROCC: how negatives are generated



- Add Gaussian noise to a normal image.
- Ascend the loss for the anomaly target.
- Project displacement to $[r, \gamma r]$.
- Train on the resulting negative.

CIFAR author code: 50 ascent steps; project every 10; $\gamma = 1$ gives a sphere.

Author CIFAR runner | Adversarial training source.

DROCC: native CIFAR result

Dataset	Variant	Paper (%)	Measured (%)
cifar10	fixed final	74.23	64.71 ± 2.18
cifar10	best test epoch	74.23	65.99 ± 2.00

- Author image CNN and Table 11 parameters; five epochs.
- All 10 classes, three seeds; 5,000 train normals / 10,000 test.
- Best-test epoch uses test labels; interpret it as optimistic.

Paper: Goyal et al., Table 1; class macro of printed means. Protocols differ.

Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

Kernel SVDD: source protocol and limits

Dataset	Variant	Paper (%)	Measured (%)
cifar10	$\nu = .01$	64.78	60.56 ± 0.20
cifar10	$\nu = .1$	64.78	60.67 ± 0.26
mnist	$\nu = .01$	91.29	95.51 ± 0.05
mnist	$\nu = .1$	91.29	95.51 ± 0.04

- Ruff image baseline: train-only PCA 95%, full kernel fit.
- Gamma tuned on 1,000 labeled test images; evaluate the other 9,000.
- Modern libsvm / Gaussian SVDD equivalence checked by independent QP.

Tax–Duin2004 original numeric table remains unreproduced: missing exact folds and search settings.

Tax author reference toolbox | Paper column: Ruff Table 1, better nu; measured: 10 seeds per nu.

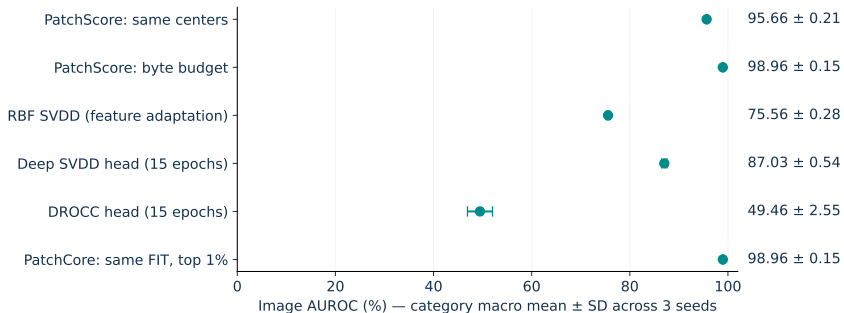
Common MVTec: one shared benchmark

Normal train	Test normal	Test anomaly
3,629	467	1,258

- 15 categories $\times$ 3 seeds; author WR50/V1 1,024-D descriptors.
- Image split: 60% FIT / 20% component calibration / remainder threshold.
- All methods: top 1% patch mean; normal-only image thresholds.
- Deep head: AE 5+SVDD 15; DROCC head: 15 epochs including 10 warmup.

Feature heads are controlled adaptations; the CNN is frozen.

Common MVTec: baseline results



PatchCore keeps a 10% coreset. PatchScore byte budget is the separate NBD memory control.

Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

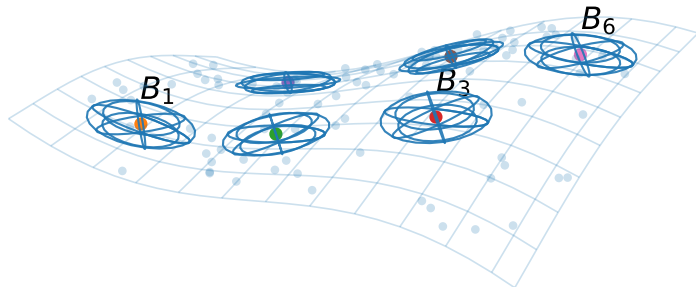
Common MVTec: operating points

Method	FPR (%)	TPR (%)
PatchScore: same centers	11.40	79.97
PatchScore: byte budget	13.71	89.30
RBF SVDD	7.27	41.69
Deep SVDD head	4.92	58.05
DROCC head	4.46	8.43
PatchCore: same FIT, top 1%	13.35	89.28
NBD: B+A+D+F	11.16	46.73

$\alpha = .05$, $k = \lceil (n + 1) \cdot .95 \rceil$; threshold = ∞ if $k > n$; strict $s > \tau$.

Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

NBD: local bubbles, one shared graph



Schematic: local ellipsoids around a curved surface.

Each bubble stores

A center, tangent variation, and normal thickness.

The graph connects them

Diffusion compares the regions each bubble can reach.

Normal features $\rightarrow$ bubbles $\rightarrow$ shared graph $\rightarrow$ diffusion distance $\rightarrow$ query score

Hypothesis tested: a normal query has local support and attaches to a coherent region of the graph.

A local normality bubble

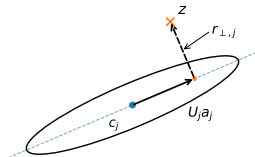
$$B_j = (c_j, U_j, \Lambda_j, \sigma_{\perp,j}^2), \quad U_j^\top U_j = I_{d_j}, \quad \Pi_j = U_j U_j^\top.$$

Local model

$$z = c_j + U_j a + \varepsilon_{\perp,j}, \quad a \perp \varepsilon_{\perp,j}, \\ a \sim \mathcal{N}(0, \Lambda_j), \quad \varepsilon_{\perp,j} \sim \mathcal{N}(0, \sigma_{\perp,j}^2 (I - \Pi_j)).$$

$$\Sigma_j = U_j \Lambda_j U_j^\top + \sigma_{\perp,j}^2 (I - \Pi_j).$$

Tangent motion + normal residual



Fit by local PCA

Neighborhood mean $\rightarrow c_j$; top d_j eigenpairs $\rightarrow U_j, \Lambda_j$.

$$\sigma_{\perp,j}^2 = \frac{1}{D - d_j} \sum_{k=d_j+1}^D \hat{\lambda}_{jk}.$$

Executed: 128 bubbles; 128 neighbors; $d_j = 16$. The same-center control uses the fitted neighborhood means c_j .

Scoring a query

$$a_j(z) = U_j^\top (z - c_j), \quad r_{\perp,j}(z) = (I - \Pi_j)(z - c_j).$$

Local energy

$$E_j(z) = \underbrace{\frac{a_j^\top (\Lambda_j + \epsilon I)^{-1} a_j}{d_j}}_{\text{tangent rarity}} + \beta \underbrace{\frac{\|r_{\perp,j}\|_2^2}{(D - d_j)(\sigma_{\perp,j}^2 + \epsilon)}}_{\text{off-manifold residual}}.$$

Absolute fit

$$S_B(z) = \min_{j \in \mathcal{C}(z)} E_j(z),$$

$$S_A(z) = -T_b \log \left[\frac{1}{K} \sum_{j \in \mathcal{C}(z)} e^{-E_j(z)/T_b} \right].$$

Attachment to the graph

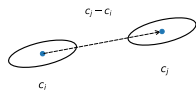
$$b_j(z) = \frac{e^{-E_j(z)/T_b}}{\sum_{k \in \mathcal{C}(z)} e^{-E_k(z)/T_b}}.$$

$K = |\mathcal{C}(z)|$ candidates; $b_j = 0$ outside.
 $\sum_j b_j(z) = 1.$

Executed: $K = 8$, $T_b = 1$, $\beta = 1$, $\epsilon = 10^{-6}$. This dimension-normalized energy is not Gaussian NLL.

Comparing two bubbles

Center displacement



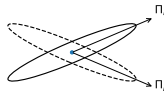
Earlier: Euclidean distance

$$r_c^E(i, j) = \|c_i - c_j\|_2^2.$$

Used here: two-way bubble energy

$$r_c^G(i, j) = \frac{1}{2} [E_i(c_j) + E_j(c_i)].$$

Frame disagreement



$$\begin{aligned} r_F(i, j) &= \|\Pi_i - \Pi_j\|_F^2 \\ &= d_i + d_j - 2 \|U_i^\top U_j\|_F^2. \end{aligned}$$

Compare tangent subspaces, not basis-vector signs.

Optional: thickness



$$r_\sigma(i, j) = \left[\log \frac{\sigma_{\perp, i}^2 + \epsilon}{\sigma_{\perp, j}^2 + \epsilon} \right]^2.$$

Not used in the reported benchmark.
Not a full covariance distance.

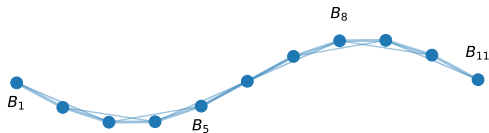
Executed graph: two-way center energy + frame disagreement. Thickness weight is zero.

The earlier Euclidean distance and the optional thickness term are shown for context, not as tested ablations.

From bubbles to a graph

Edge dissimilarity, before diffusion

$$\delta_B(i,j) = \frac{r_c(i,j)}{\tau_c} + \lambda_F \frac{r_F(i,j)}{\tau_F}.$$



Toy graph. Each node is one bubble; thicker edges have larger affinity.

Affinity and transition

For symmetric local candidate edges $\mathcal{E}$,

$$W_{ij} = \mathbf{1}_{\{(i,j) \in \mathcal{E}\}} e^{-\delta_B(i,j)} \quad (i \neq j),$$

$$W_{ii} = \eta > 0, \quad s_i = \sum_j W_{ij},$$

$$P_{ij} = \frac{W_{ij}}{s_i}, \quad \pi_i = \frac{s_i}{\sum_m s_m}.$$

Fit the graph once on normal data. Every query uses the same bubble nodes.

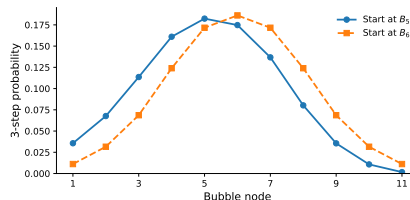
Executed: 12 graph neighbors; $\lambda_F = 1$, $\eta = 1$. No thickness penalty. δ_B need not be a metric.

OCC REPRODUCTION

Diffusion distance

Similar t -step transition distributions give a smaller diffusion distance:

$$D_t^2(i, j) = \sum_m \frac{[(P^t)_{im} - (P^t)_{jm}]^2}{\pi_m}.$$



Compute with eigenvectors

$$P\psi_\ell = \lambda_\ell \psi_\ell, \quad \sum_i \pi_i \psi_\ell(i) \psi_k(i) = \delta_{\ell k}.$$

$$D_t^2(i, j) = \sum_{\ell=1}^{M-1} \lambda_\ell^{2t} [\psi_\ell(i) - \psi_\ell(j)]^2.$$

$$\Phi_t(B_j) = (\lambda_1^t \psi_1(j), \dots, \lambda_L^t \psi_L(j)),$$

$$D_{t,L}^2(i, j) = \|\Phi_t(B_i) - \Phi_t(B_j)\|^2.$$

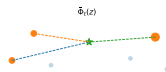
[4] Coifman–Lafon (2006). Executed: all $L = M - 1$ nonconstant modes, $\mathcal{T} = \{1, 3, 5\}$. The figure is schematic.

Does the query attach to one region?



Nearby bubbles: lower dispersion

Schematic diffusion coordinates. Circle area: $b_j(z)$. Star: weighted mean.



Distant bubbles: higher dispersion

$$S_D(z) = \frac{1}{2|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{i,j} b_i(z) b_j(z) D_{t,L}^2(i,j).$$

Equivalent variance at time t

$$\bar{\Phi}_t(z) = \sum_j b_j(z) \Phi_t(B_j),$$

$$S_{D,t}(z) = \sum_j b_j(z) \left\| \Phi_t(B_j) - \bar{\Phi}_t(z) \right\|^2.$$

Frame inconsistency

$$S_F(z) = \frac{1}{2} \sum_{i,j} b_i(z) b_j(z) r_F(i,j).$$

Executed: $\mathcal{T} = \{1, 3, 5\}$.

A one-bubble outlier can have $S_D \approx 0$; S_B and S_A still measure its absolute mismatch.

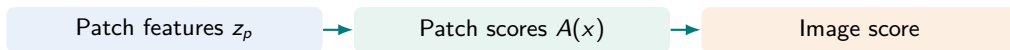
NBD: component scaling and final score

Local fit + diffusion and frame consistency

$$S_{\text{NBD}}(z) = \tilde{S}_B(z) + \tilde{S}_A(z) + \tilde{S}_D(z) + \lambda_F \tilde{S}_F(z).$$

NBD only: $\mathcal{Z}_{\text{cal}}$ contains patches from the 721 reserved normal images:

$$\tilde{S}_q(z) = -\log \left[\frac{1 + \sum_{u \in \mathcal{Z}_{\text{cal}}} \mathbf{1}_{\{S_q(u) \geq S_q(z)\}}}{|\mathcal{Z}_{\text{cal}}| + 1} \right], \quad q \in \{B, A, D, F\}.$$



$$A(x)_p = S_{\text{NBD}}(z_p).$$

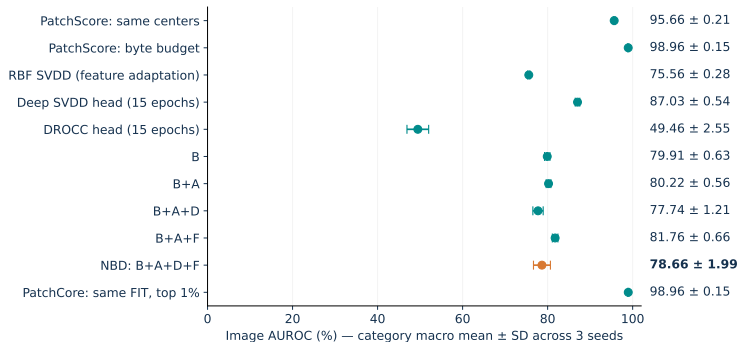
$$S_{\text{image}}(x) = \frac{1}{|\mathcal{P}_\rho(x)|} \sum_{p \in \mathcal{P}_\rho(x)} A(x)_p.$$

$\mathcal{P}_\rho$: top 1% (8 of 784 patches). $\lambda_F = 1$. This is not a calibrated anomaly probability.

Component scaling uses the reserved subset, not the 734 threshold images.

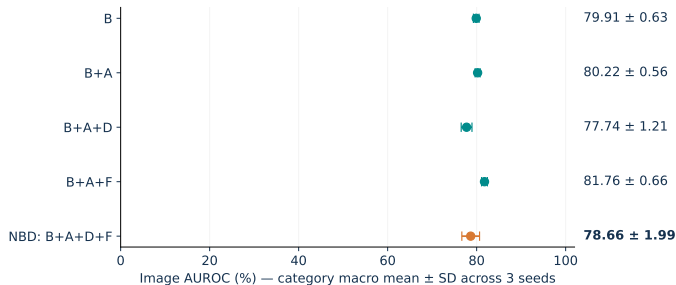
After aggregation, NBD uses the same image-threshold rule as the baselines.

NBD versus common baselines



Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

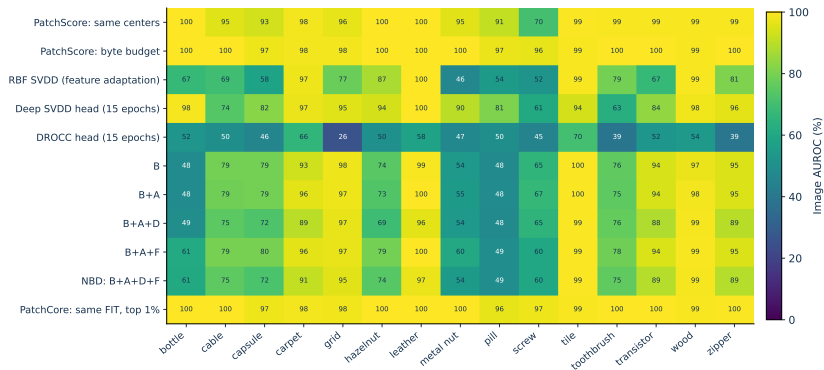
Ablations: keep the declared NBD result



B: bubble fit; *A*: support; *D*: diffusion; *F*: frame. Final NBD stays $B + A + D + F$.

Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

All 15 categories, all 11 variants



Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

Complete common results

Method	Image AUROC (%)	AP (%)
PatchScore: same centers	95.66 $\pm$ 0.21	98.41 $\pm$ 0.10
PatchScore: byte budget	98.96 $\pm$ 0.15	99.70 $\pm$ 0.05
RBF SVDD (feature adaptation)	75.56 $\pm$ 0.28	88.92 $\pm$ 0.19
Deep SVDD head (15 epochs)	87.03 $\pm$ 0.54	94.12 $\pm$ 0.33
DROCC head (15 epochs)	49.46 $\pm$ 2.55	72.88 $\pm$ 1.44
B	79.91 $\pm$ 0.63	91.99 $\pm$ 0.88
B+A	80.22 $\pm$ 0.56	92.20 $\pm$ 0.68
B+A+D	77.74 $\pm$ 1.21	90.65 $\pm$ 0.22
B+A+F	81.76 $\pm$ 0.66	92.75 $\pm$ 0.77
NBD: B+A+D+F	78.66 $\pm$ 1.99	90.99 $\pm$ 0.77
PatchCore: same FIT, top 1%	98.96 $\pm$ 0.15	99.71 $\pm$ 0.04

Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

Coverage and reproducibility

Track	Completed / planned result groups
DROCC	60/60
DeepSVDD	400/400
Gaussian_OCSVM_equiv_SVDD	400/400
common_mvtec	90/90
native_patchcore	45/45

- Independent image-score audit: 8,262,100 exported predictions.
- Pinned sources, configs, split IDs, checkpoints, replay and raw CSVs retained.
- Original full schedules and historical v2 results remain available.

Measured rerun: OCC / reproduction-2026-09-14. See CSVs and source audit.

Inference: shared CNN and scorer costs

Method	Median ms	State MiB
Shared author WR50 descriptor	30.91	263.03
PatchScore: same centers	1.30	0.50
PatchScore: byte budget	1.47	9.01
RBF SVDD	10.82	16.02
Deep SVDD head	1.41	0.25
DROCC head	1.50	0.25
NBD: B+A+D+F	7.47	9.13
PatchCore: same FIT, top 1%	1.91	38.28

One bottle image, seed 0; batch 1; 30 repeats. NBD calibration arrays are additional; see report.

CUDA-synchronized after training. Scorer time excludes shared CNN; state is not peak GPU memory.

Appendix: historical v2 shared-feature results

Historical method	Image AUROC (%)
PatchScore: same centers	95.86 $\pm$ 0.23
PatchScore: byte budget	98.56 $\pm$ 0.21
RBF SVDD (feature adaptation)	78.20 $\pm$ 0.26
Deep SVDD head (100 epochs)	81.85 $\pm$ 0.52
DROCC head (100 epochs)	54.45 $\pm$ 5.43
B	82.70 $\pm$ 1.81
B+A	82.80 $\pm$ 1.62
B+A+D	79.72 $\pm$ 2.46
B+A+F	82.82 $\pm$ 1.88
NBD: B+A+D+F	78.79 $\pm$ 2.37

Previous 1,536-D extractor, AE 50 / heads 100 epochs. Preserved evidence; a different protocol.

Historical v2 at 91223d6. Do not infer a causal feature effect from this comparison.

References and evidence

- Ruff et al.: Deep One-Class Classification (2018).
- Goyal et al.: DROCC (2020).
- Roth et al.: PatchCore (2022), released author code.
- Tax and Duin: Support Vector Data Description (2004).
- Coifman and Lafon: Diffusion Maps (2006).
- Code, full tables, handoff and source audit: [dathuynh1108/OCC](#).

Scores describe these declared protocols and training budgets.